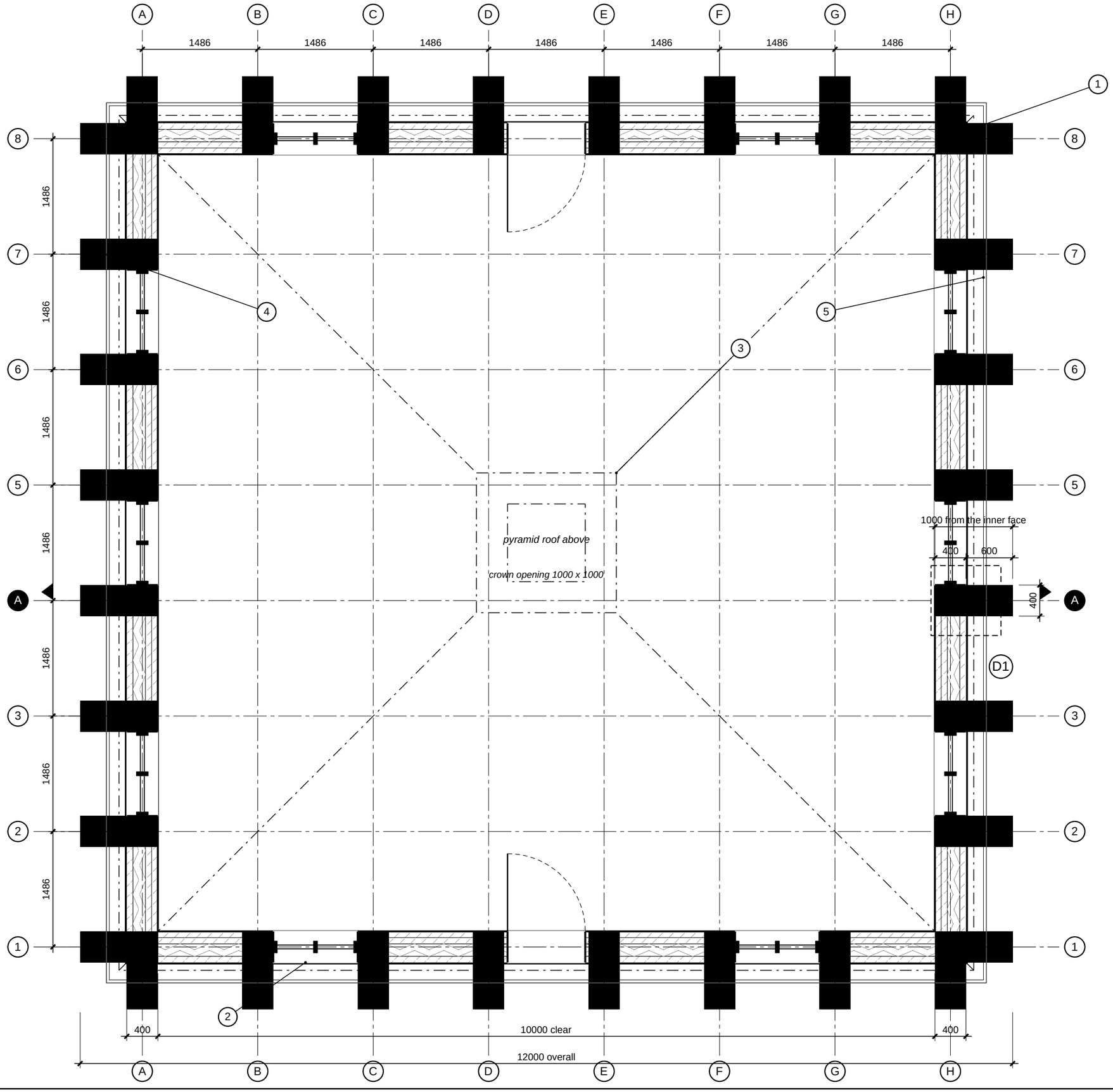


Floor plan

10 x 10 m clear · buttress every 1.5 m · printed walls

1. Printed wall 400 (= roof base width). Buttresses solid black: 400 wide (same as the wall), 1000 deep from the inner face (flush with the wall inside), so they project 600 outside; one about every 1.5 m (7 bays of 1486 on the wall centreline, 8 per side, corners included).
2. Every other bay (the 2nd, 4th and 6th) is a full-height curtain wall, the whole bay between two buttresses; the middle bay of the south and north walls has a 1000 door instead. The other bays stay printed wall. Positions and sizes are placeholders.
3. Pyramid roof above (dashed): 1000 x 1000 clear crown opening, hips to the corners; the roof base is 400 thick square to the slope, so the shell ends 100 outside the wall face, and 262 with the insulation and outer layer.
4. Buttress axes 1486 apart along each wall (centre to centre), corners included.
5. Roof build-up outside the shell (outline at the eave): insulation 100 (very thin line), then the outer layer 30.

Notes

Section A-A cuts through the 4th buttress from the south on the east and west walls (y = -743).

Detail D1 is the eave at the east buttress on that cut.

0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP
Scale 1:50 at A3 · dimensions in mm
2026-10-14 · hand estimates, not engineered

01

Floor plan · roof grid A

10 x 10 m clear · buttress every 1.5 m · printed walls

1. Printed wall 400 (= roof base width). Buttresses solid black: 400 wide (same as the wall), 1000 deep from the inner face (flush with the wall inside), so they project 600 outside; one about every 1.5 m (7 bays of 1486 on the wall centreline, 8 per side, corners included).
2. Every other bay (the 2nd, 4th and 6th) is a full-height curtain wall, the whole bay between two buttresses; the middle bay of the south and north walls has a 1000 door instead. The other bays stay printed wall. Positions and sizes are placeholders.
3. Pyramid roof above (dashed): 1000 x 1000 clear crown opening, hips to the corners; the roof base is 400 thick square to the slope, so the shell ends 100 outside the wall face, and 262 with the insulation and outer layer.
4. Buttress axes 1486 apart along each wall (centre to centre), corners included.
5. Roof build-up outside the shell (outline at the eave): insulation 100 (very thin line), then the outer layer 30.
6. Cast ribs of roof grid A seen from above (dashed), with the bottom ring (the 300 solid base) and the horizontal / top rings.

Notes

Section A-A cuts through the 4th buttress from the south on the east and west walls (y = -743).

Detail D1 is the eave at the east buttress on that cut.

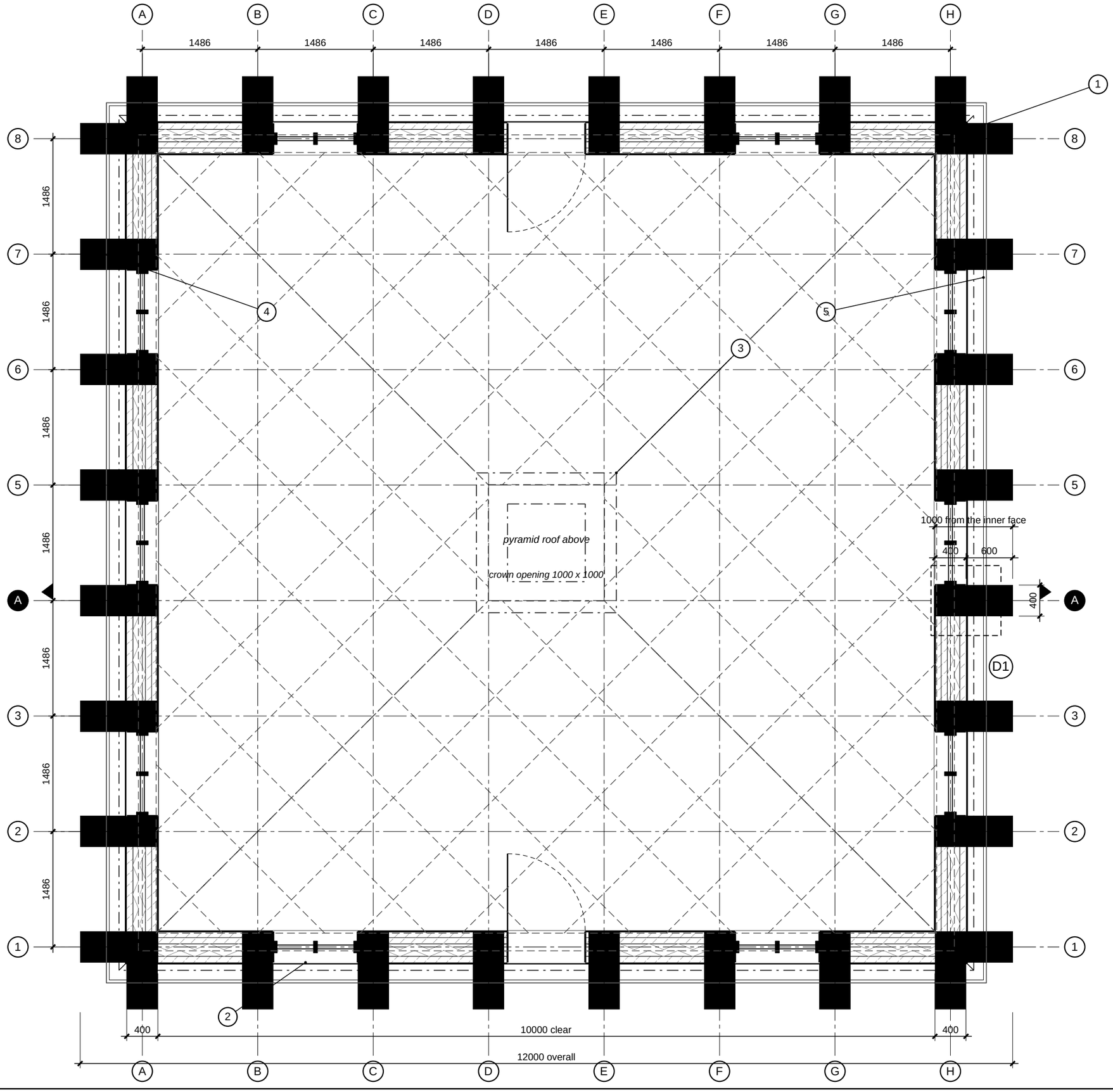
0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP

Scale 1:50 at A3 · dimensions in mm

2026-10-14 · hand estimates, not engineered

02



Floor plan · roof grid B

10 x 10 m clear · buttress every 1.5 m · printed walls

1. Printed wall 400 (= roof base width). Buttresses solid black: 400 wide (same as the wall), 1000 deep from the inner face (flush with the wall inside), so they project 600 outside; one about every 1.5 m (7 bays of 1486 on the wall centreline, 8 per side, corners included).
2. Every other bay (the 2nd, 4th and 6th) is a full-height curtain wall, the whole bay between two buttresses; the middle bay of the south and north walls has a 1000 door instead. The other bays stay printed wall. Positions and sizes are placeholders.
3. Pyramid roof above (dashed): 1000 x 1000 clear crown opening, hips to the corners; the roof base is 400 thick square to the slope, so the shell ends 100 outside the wall face, and 262 with the insulation and outer layer.
4. Buttress axes 1486 apart along each wall (centre to centre), corners included.
5. Roof build-up outside the shell (outline at the eave): insulation 100 (very thin line), then the outer layer 30.
6. Cast ribs of roof grid B seen from above (dashed), with the bottom ring (the 300 solid base) and the horizontal / top rings.

Notes

Section A-A cuts through the 4th buttress from the south on the east and west walls (y = -743).

Detail D1 is the eave at the east buttress on that cut.

0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP

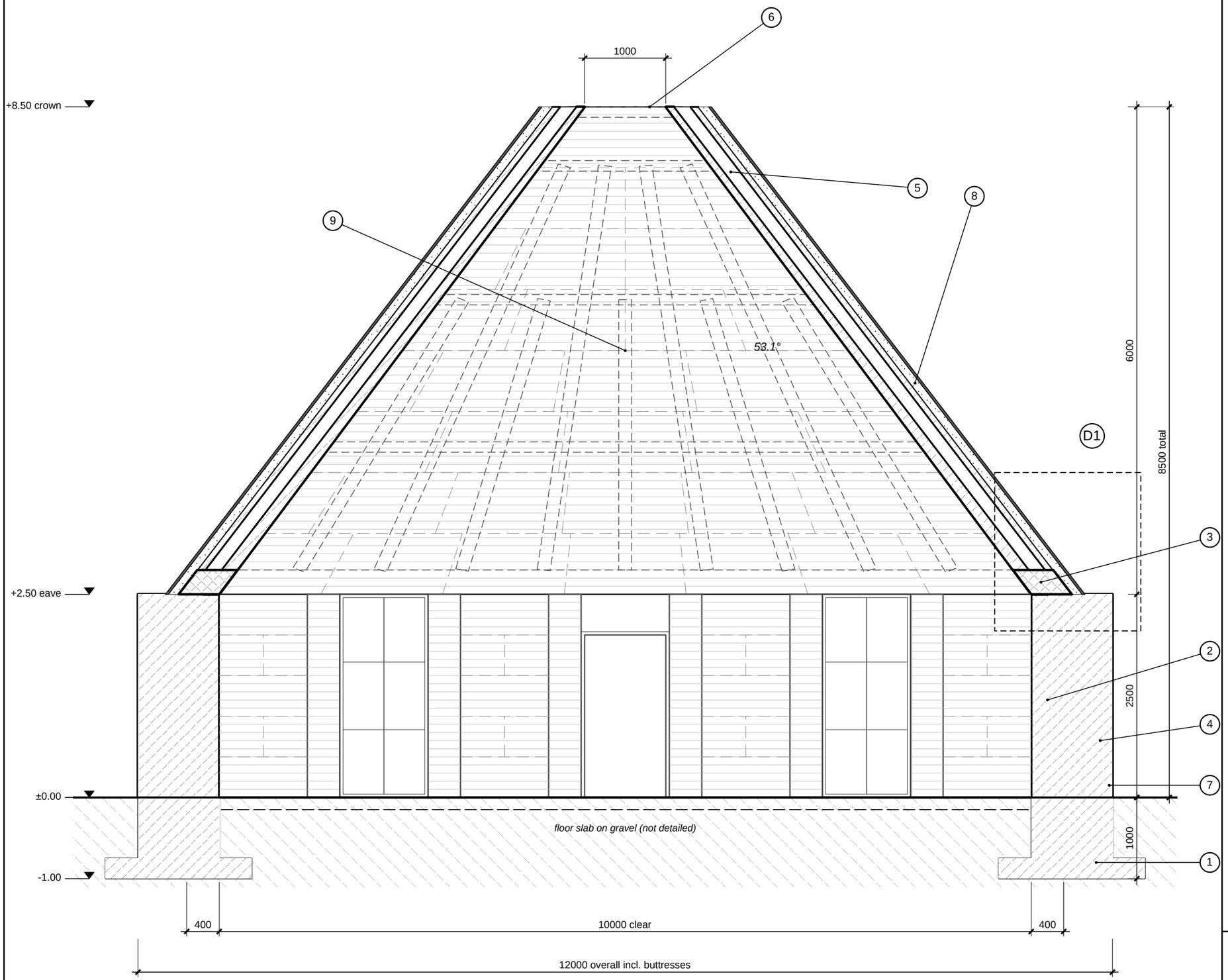
Scale 1:50 at A3 · dimensions in mm

2026-10-14 · hand estimates, not engineered

03

Total section A-A · grid A

through the east and west buttresses, looking north



1. Footing 1800 x 250, underside at -1000 (frost depth and bearing are guesses).
2. Wall 400. The cut passes through a buttress, so the wall is solid with the buttress here (no insulation). Between buttresses it is printed like the roof: two 80 inner layers, a 160 air / insulation zone, an 80 outer layer.
3. Solid base of the roof: the first 300 (vertical) above the eave is completely solid concrete, 400 wide, no air zone; there is no ring beam.
4. Buttress 400 wide, 1000 from the inner face (600 beyond the wall), one about every 1.5 m (8 per side); it rises to +2.50, flush with the roof base.
5. Roof shell: 400 thick square to the slope at the base (two 80 inner layers, a 160 air zone, an 80 outer layer), 320 at the crown; about 500 / 400 horizontally.
6. Crown opening 1000 clear; roof rises 6000 over 4500, slope 53.1° (36.9° from the vertical).
7. Everything is 3D printed in 10 mm layers; crossed hatch = solid cast concrete (roof base, ribs). Printed layer joints and the cast fill of the buttress and footing are shown in the roof-grid sheets and in detail D1.
8. Outside the printed shell: insulation 100 (dotted), then the outer layer (cladding 30, dark).
9. Cast ribs of the roof grid, seen on the back face as dashed outlines (roof grid A, ruled-surface columns). The cut lies in the air zone between ribs; rib sections are on the roof-grid sheets.

Notes

At the crown the second inner layer ends, so the leaf is 80 there.

The cut runs through the 4th buttress from the south (y = -743), as on the floor plan. Beyond it the north wall is seen as on the plan: buttress every 1.5 m (strips outside the wall), every other bay printed wall, the bays in between full-height curtain wall (door in the middle one).

Faint grid on the back roof face and the north wall = printed pieces in running bond, about 1500 wide (assumed sizes). Dashed outlines on the back roof face = cast ribs.

0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP

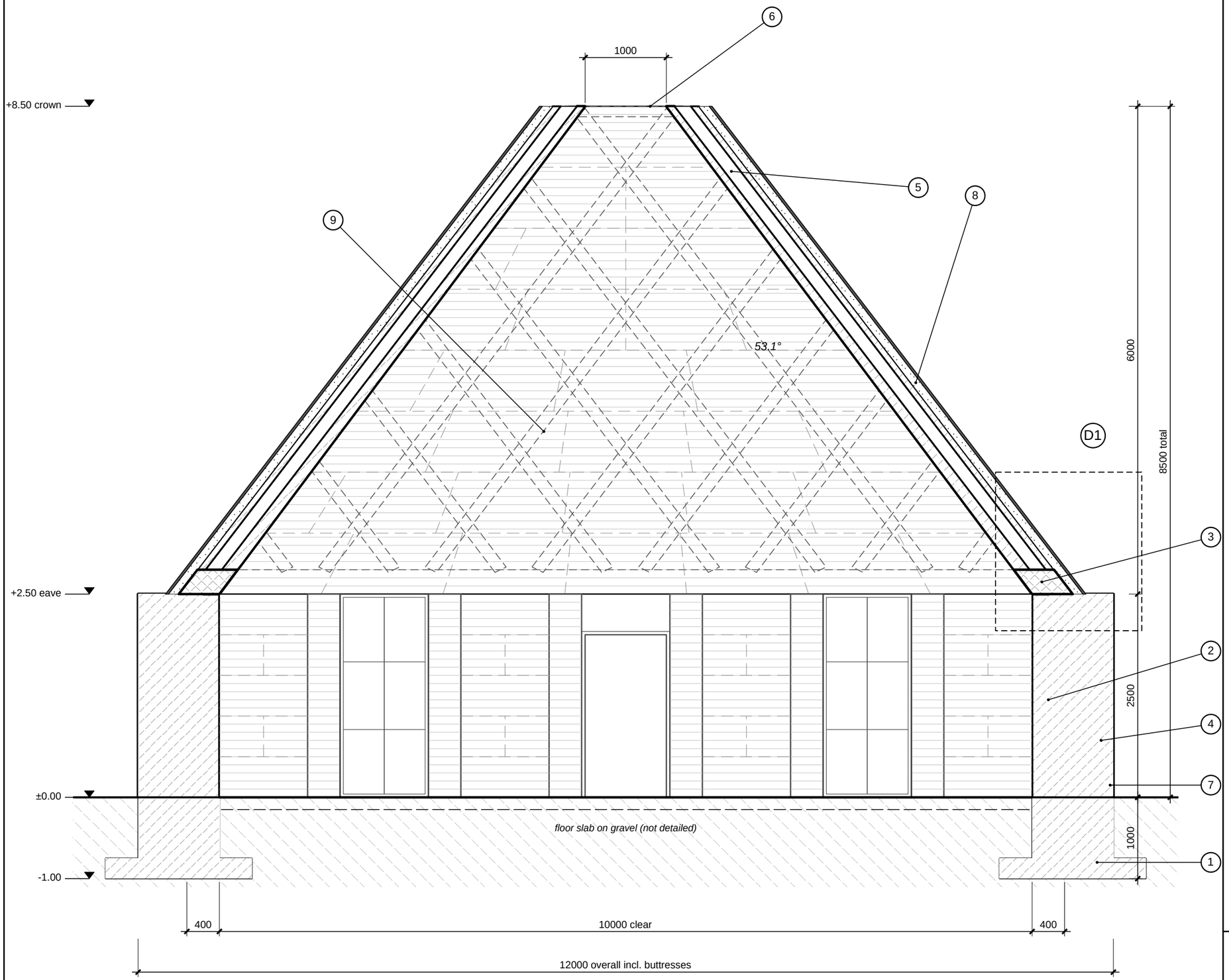
Scale 1:50 at A3 · dimensions in mm

2026-10-14 · hand estimates, not engineered

04

Total section A-A · grid B

through the east and west buttresses, looking north



1. Footing 1800 x 250, underside at -1000 (frost depth and bearing are guesses).
2. Wall 400. The cut passes through a buttress, so the wall is solid with the buttress here (no insulation). Between buttresses it is printed like the roof: two 80 inner layers, a 160 air / insulation zone, an 80 outer layer.
3. Solid base of the roof: the first 300 (vertical) above the eave is completely solid concrete, 400 wide, no air zone; there is no ring beam.
4. Buttress 400 wide, 1000 from the inner face (600 beyond the wall), one about every 1.5 m (8 per side); it rises to +2.50, flush with the roof base.
5. Roof shell: 400 thick square to the slope at the base (two 80 inner layers, a 160 air zone, an 80 outer layer), 320 at the crown; about 500 / 400 horizontally.
6. Crown opening 1000 clear; roof rises 6000 over 4500, slope 53.1° (36.9° from the vertical).
7. Everything is 3D printed in 10 mm layers; crossed hatch = solid cast concrete (roof base, ribs). Printed layer joints and the cast fill of the buttress and footing are shown in the roof-grid sheets and in detail D1.
8. Outside the printed shell: insulation 100 (dotted), then the outer layer (cladding 30, dark).
9. Cast ribs of the roof grid, seen on the back face as dashed outlines (roof grid B, diamond lattice). The cut lies in the air zone between ribs; rib sections are on the roof-grid sheets.

Notes

At the crown the second inner layer ends, so the leaf is 80 there.

The cut runs through the 4th buttress from the south (y = -743), as on the floor plan. Beyond it the north wall is seen as on the plan: buttress every 1.5 m (strips outside the wall), every other bay printed wall, the bays in between full-height curtain wall (door in the middle one).

Faint grid on the back roof face and the north wall = printed pieces in running bond, about 1500 wide (assumed sizes). Dashed outlines on the back roof face = cast ribs.

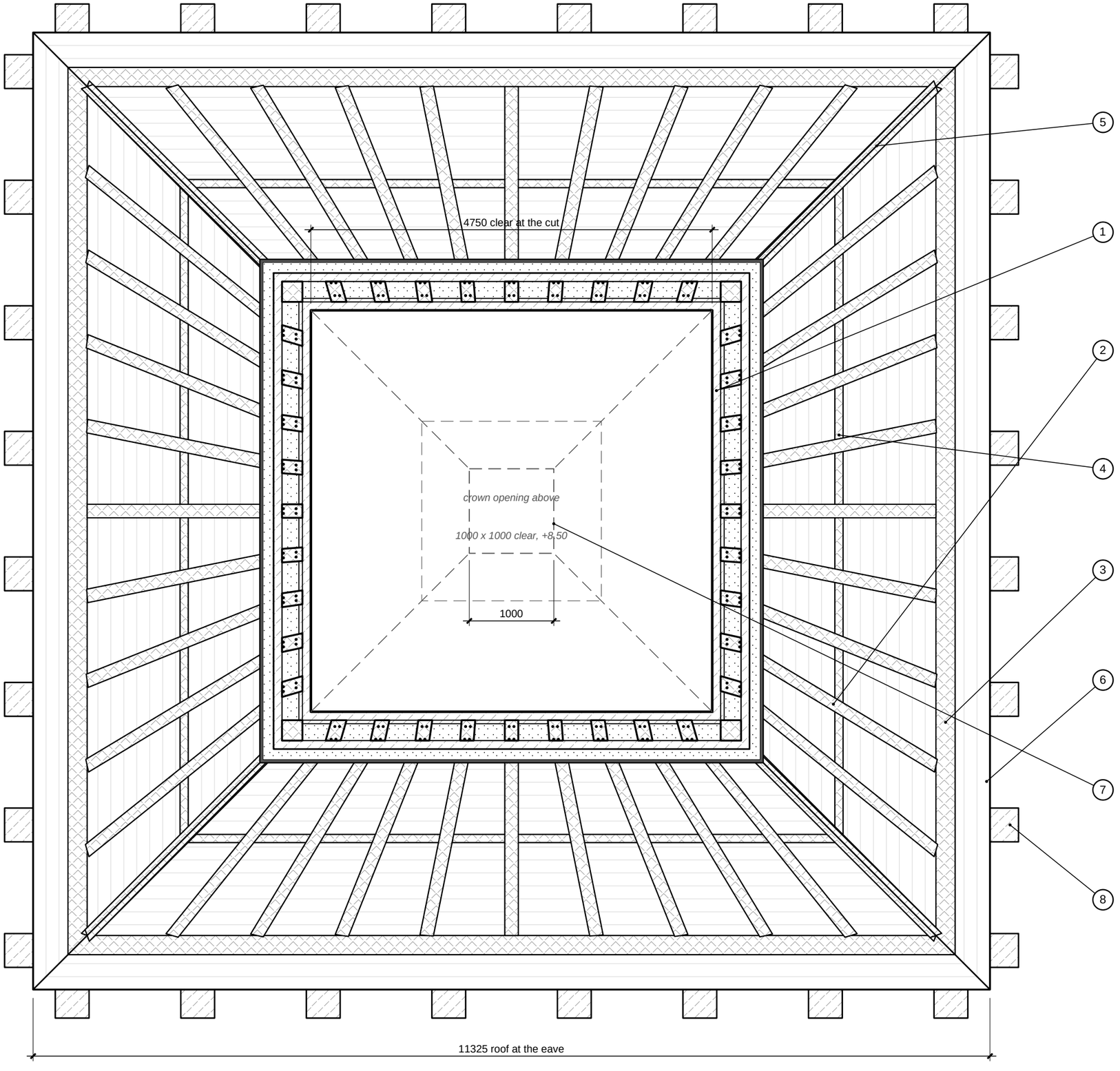
0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP

Scale 1:50 at A3 · dimensions in mm

2026-10-14 · hand estimates, not engineered

05



Roof plan, cut at +6.00 · grid A

horizontal cut through the roof, looking down on the ribs

1. Cut at +6.00 (3500 above the eave, just below the horizontal rib at +6.13): printed layers, air zone, cast ribs cut (crossed hatch), outer layer, insulation 100 and outer layer 30.
2. Ribs of grid A seen below the cut: all 11 columns per face run from the base ring at the eave up to the cut (every other one stops at +6.13, just above it).
3. Base ring: the 300 solid base at the eave (+2.50 to +2.80).
4. Horizontal rib of grid A at +4.31 (the next ones, at +6.13 and +7.77, are above the cut).
5. Hips: ribs from the corners of the eave up to the cut; dashed above it, up to the crown opening.
6. Roof edge at the eave (outside of the 30 outer layer). Very faint squares = printed layer joints (real layers 10, drawn every 200 in height).
7. Above the cut (removed, dashed): hips and the crown opening at +8.50, 1000 clear inside (1400 at the middle of the shell, 2125 outside the outer layer).
8. Buttresses below the roof: their ends show beyond the roof edge; the wall under the roof is dashed.

Notes

Matches the roof-grid elevations: same rib geometry, same levels (+4.31, +6.13, +7.77 for the horizontal ribs of grid A).

The ribs lie inside the shell, between the printed inner layers and the outer layer; here they are drawn as if the outer layer, insulation and cladding were transparent.

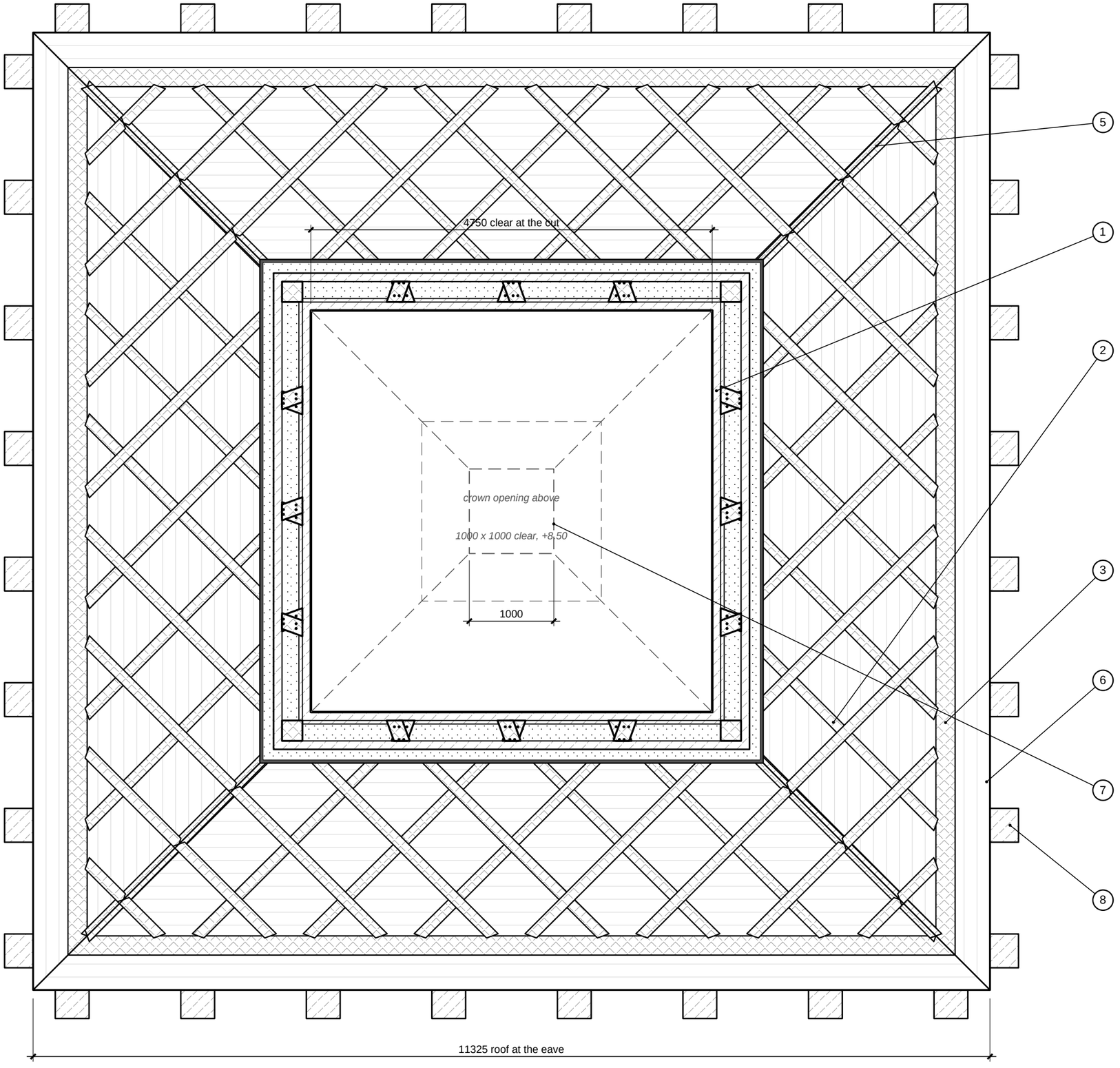
0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP

Scale 1:50 at A3 · dimensions in mm

2026-10-14 · hand estimates, not engineered

06



Roof plan, cut at +6.00 · grid B

horizontal cut through the roof, looking down on the ribs

1. Cut at +6.00 (3500 above the eave, just below the horizontal rib at +6.13): printed layers, air zone, cast ribs cut (crossed hatch), outer layer, insulation 100 and outer layer 30.
2. Ribs of grid B seen below the cut: the two families of diagonals from the base ring at the eave up to the cut and the hips.
3. Base ring: the 300 solid base at the eave (+2.50 to +2.80).
4. No horizontal ribs in grid B between the base ring and the top ring.
5. Hips: ribs from the corners of the eave up to the cut; dashed above it, up to the crown opening.
6. Roof edge at the eave (outside of the 30 outer layer). Very faint squares = printed layer joints (real layers 10, drawn every 200 in height).
7. Above the cut (removed, dashed): hips and the crown opening at +8.50, 1000 clear inside (1400 at the middle of the shell, 2125 outside the outer layer).
8. Buttresses below the roof: their ends show beyond the roof edge; the wall under the roof is dashed.

Notes

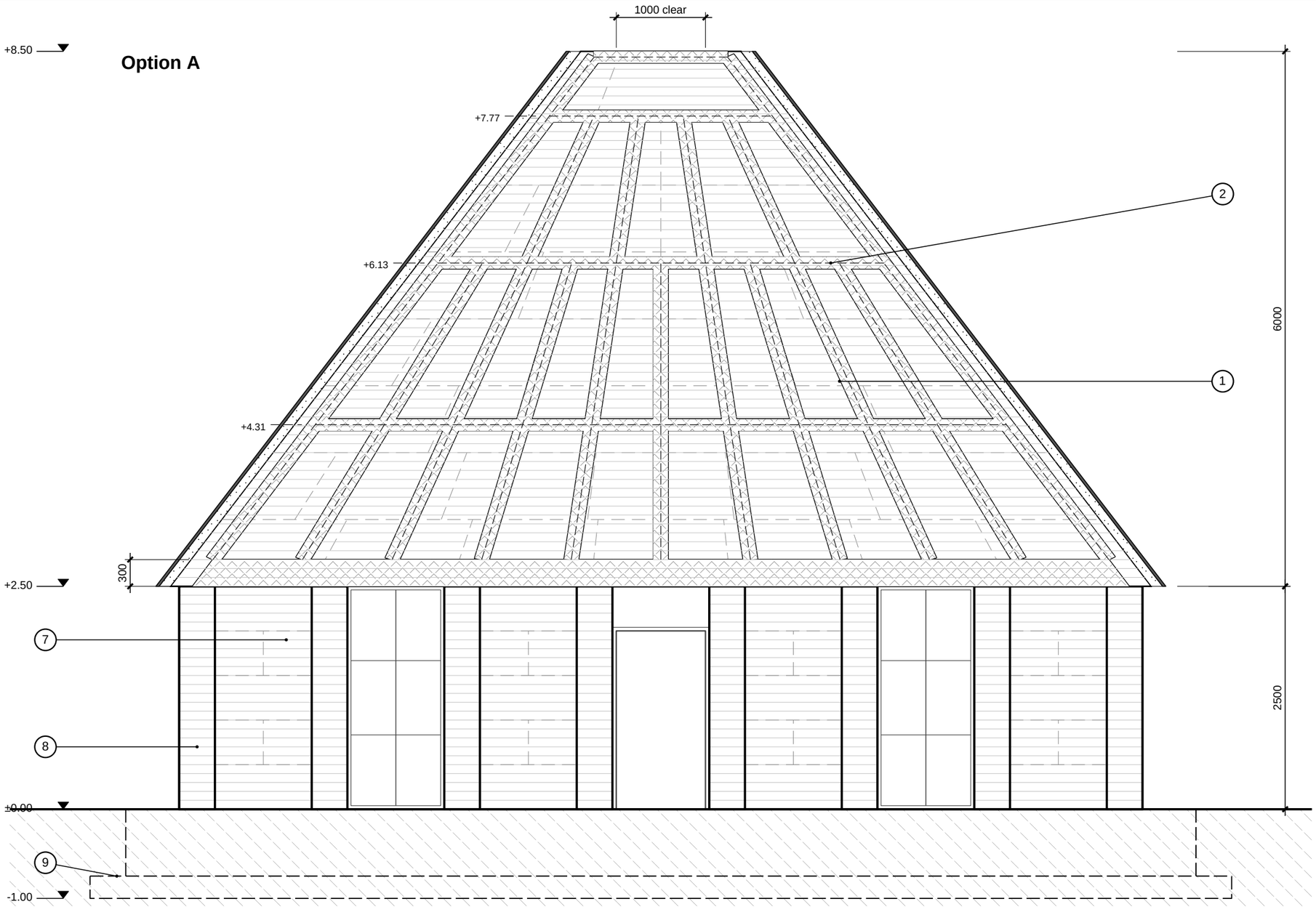
Matches the roof-grid elevations: same rib geometry, same levels (+4.31, +6.13, +7.77 for the horizontal ribs of grid A).

The ribs lie inside the shell, between the printed inner layers and the outer layer; here they are drawn as if the outer layer, insulation and cladding were transparent.

0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP
Scale 1:50 at A3 · dimensions in mm
2026-10-14 · hand estimates, not engineered

07



Roof grid A: ruled-surface columns

one roof face seen square on

1. Columns run in straight lines from the base edge (every 1050) towards the crown edge, so they converge like the ruled surface of the sketch.
2. When the spacing drops to 500 (about +6.13) every other column stops; at about +7.77 only the four hip columns go on to the crown ring.
3. Rib (column): a channel 160 wide x 240 deep, printed in the 160 air zone + layer 2 stays in place as formwork and is filled with cast concrete, 4 Ø16 + Ø8 stirrups every 200 (assumed). Fibre concrete could replace the bars (see notes).
4. Inner layer 1, 80: continuous, covers the grid on the room side.
5. Outer skin 80: continuous, covers the grid on the outside (cladding on it).
6. Between the ribs the air zone stays open for thermal insulation (dotted).
7. Printed wall 400 (same layers as the roof), to +2.50.
8. Buttresses 400 wide, 1000 from the inner face (600 outside), about every 1.5 m (7 bays of 1486 on the wall centreline, 8 per side, corners included); every other bay is printed wall, the bays between are full-height curtain wall (door in the middle one).
9. Strip footing 1800 x 250, underside at -1.00 (hidden, dashed): printed edge formwork, cast core.
10. Printed channel walls, 80 each side of the rib: permanent formwork, printed in the same pass as the shell (layer 2 turns into them).
11. Poured core: the solid base (300, full concrete over the whole 400) and the rib are poured together, in lifts of about 600 from the eave (assumed), with no joint between base and rib; dashed lines = lift joints. Printed layer joints are every 10 (drawn every 20 here).
12. Build-up outside the printed shell, seen at the sides of the roof: insulation 100 (type to be chosen), then the outer layer (cladding 30). The 160 air zone inside the shell stays open between the ribs.

Notes

Columns stop on a horizontal rib (at 3626 and 5275 above the eave); bars are lapped / hooked into it.

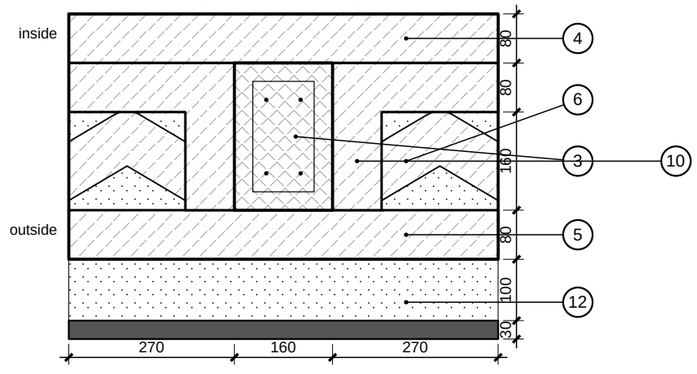
Horizontal ribs at 1813, 3626, 5275, plus the 300 solid base (bottom ring) and the top ring at the crown.

Dashed lines = printed pieces in running bond (about 1500 x 750); very faint lines = printed layer joints (10 mm layers, drawn every 100).

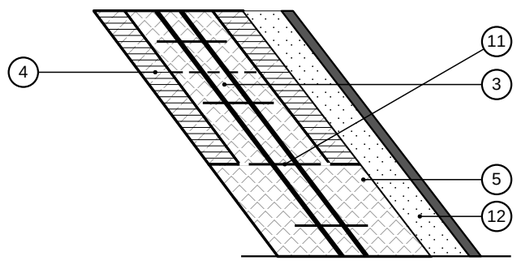
Printed mortar beads are permanent formwork; ribs, base and buttress cores are poured concrete (after the Obayashi 3dpod paper, 2023). Steel-fibre concrete could replace the rib bars, to be confirmed by the engineer.

Rough quantities, four faces: ribs 325 m, steel 2049 kg (4 Ø16 per rib), poured concrete 12.5 m3.

Plan cut through a rib at one layer, 1:10



Section along a rib at the eave, 1:20



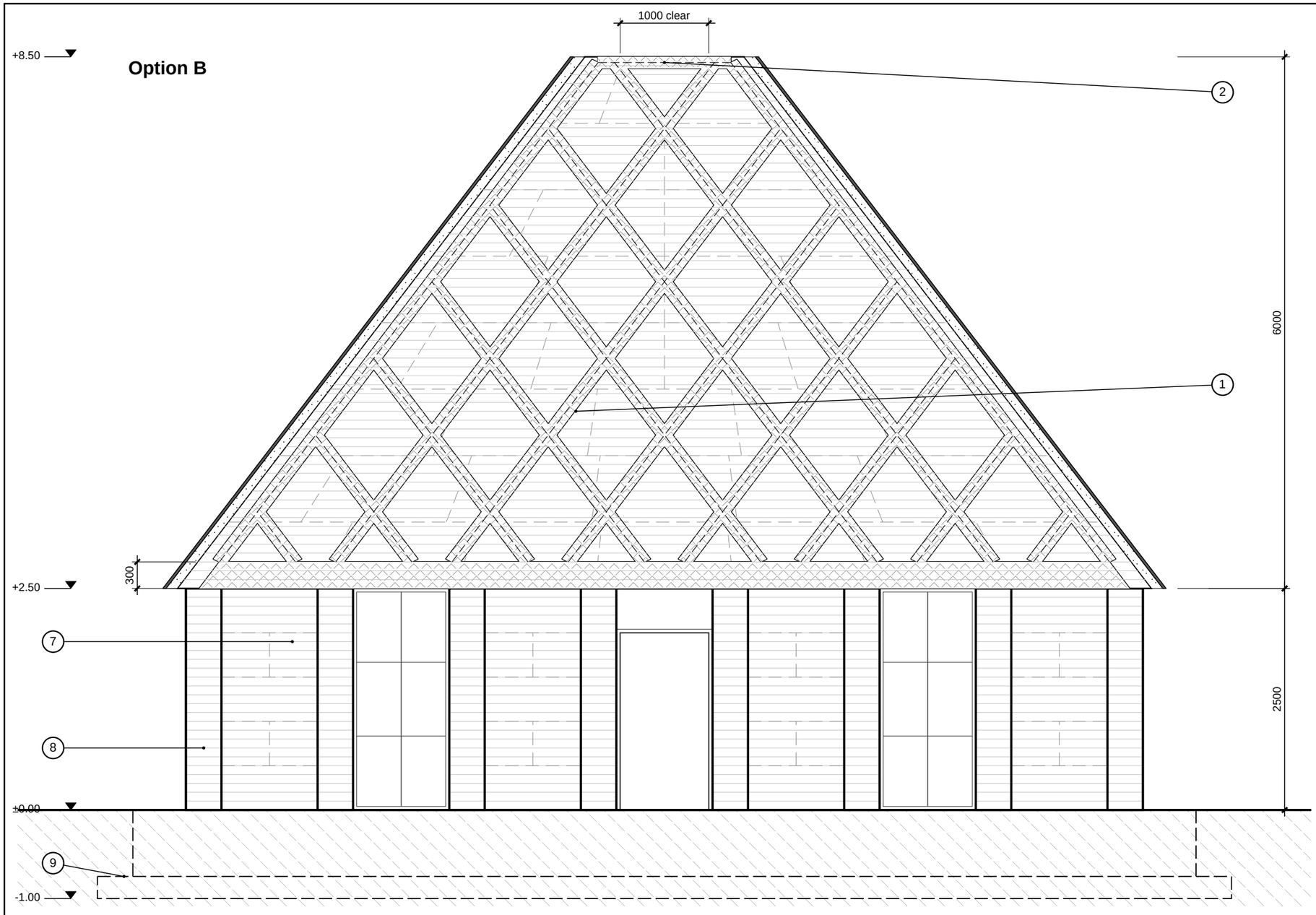
0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP

Scale 1:50 at A3 · rib plan cut 1:10, rib section 1:20 · dimensions in mm

2026-10-14 · hand estimates, not engineered

08



Roof grid B: diamond lattice

one roof face seen square on

1. Diagonals parallel to the two hips (same inclination as the roof edges, 58.9° in the face), every 1312.5 along the base: equal diamonds 1312.5 wide.
2. Bottom ring = the 300 solid base; top ring at the crown. The hips are two of the diagonals. No other horizontal ribs.
3. Rib (column): a channel 160 wide x 240 deep, printed in the 160 air zone + layer 2 stays in place as formwork and is filled with cast concrete, 4 Ø16 + Ø8 stirrups every 200 (assumed). Fibre concrete could replace the bars (see notes).
4. Inner layer 1, 80: continuous, covers the grid on the room side.
5. Outer skin 80: continuous, covers the grid on the outside (cladding on it).
6. Between the ribs the air zone stays open for thermal insulation (dotted).
7. Printed wall 400 (same layers as the roof), to +2.50.
8. Buttresses 400 wide, 1000 from the inner face (600 outside), about every 1.5 m (7 bays of 1486 on the wall centreline, 8 per side, corners included); every other bay is printed wall, the bays between are full-height curtain wall (door in the middle one).
9. Strip footing 1800 x 250, underside at -1.00 (hidden, dashed): printed edge formwork, cast core.
10. Printed channel walls, 80 each side of the rib: permanent formwork, printed in the same pass as the shell (layer 2 turns into them).
11. Poured core: the solid base (300, full concrete over the whole 400) and the rib are poured together, in lifts of about 600 from the eave (assumed), with no joint between base and rib; dashed lines = lift joints. Printed layer joints are every 10 (drawn every 20 here).
12. Build-up outside the printed shell, seen at the sides of the roof: insulation 100 (type to be chosen), then the outer layer (cladding 30). The 160 air zone inside the shell stays open between the ribs.

Notes

Every diagonal ends on a ring or on a hip, so no bar has a free end in the shell.

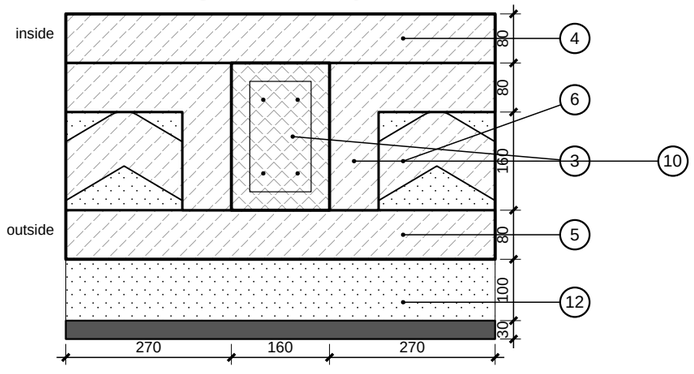
Diagonals do not stop as the roof narrows: the diamonds just get clipped by the hips.

Dashed lines = printed pieces in running bond (about 1500 x 750); very faint lines = printed layer joints (10 mm layers, drawn every 100).

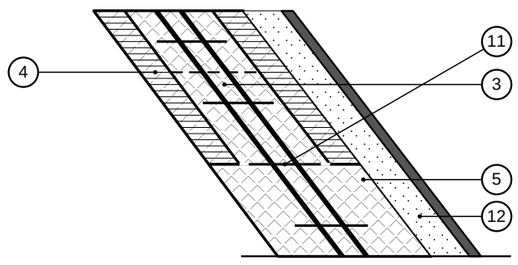
Printed mortar beads are permanent formwork; ribs, base and buttress cores are poured concrete (after the Obayashi 3dpod paper, 2023). Steel-fibre concrete could replace the rib bars, to be confirmed by the engineer.

Rough quantities, four faces: ribs 330 m, steel 2085 kg (4 Ø16 per rib), poured concrete 12.7 m3.

Plan cut through a rib at one layer, 1:10



Section along a rib at the eave, 1:20



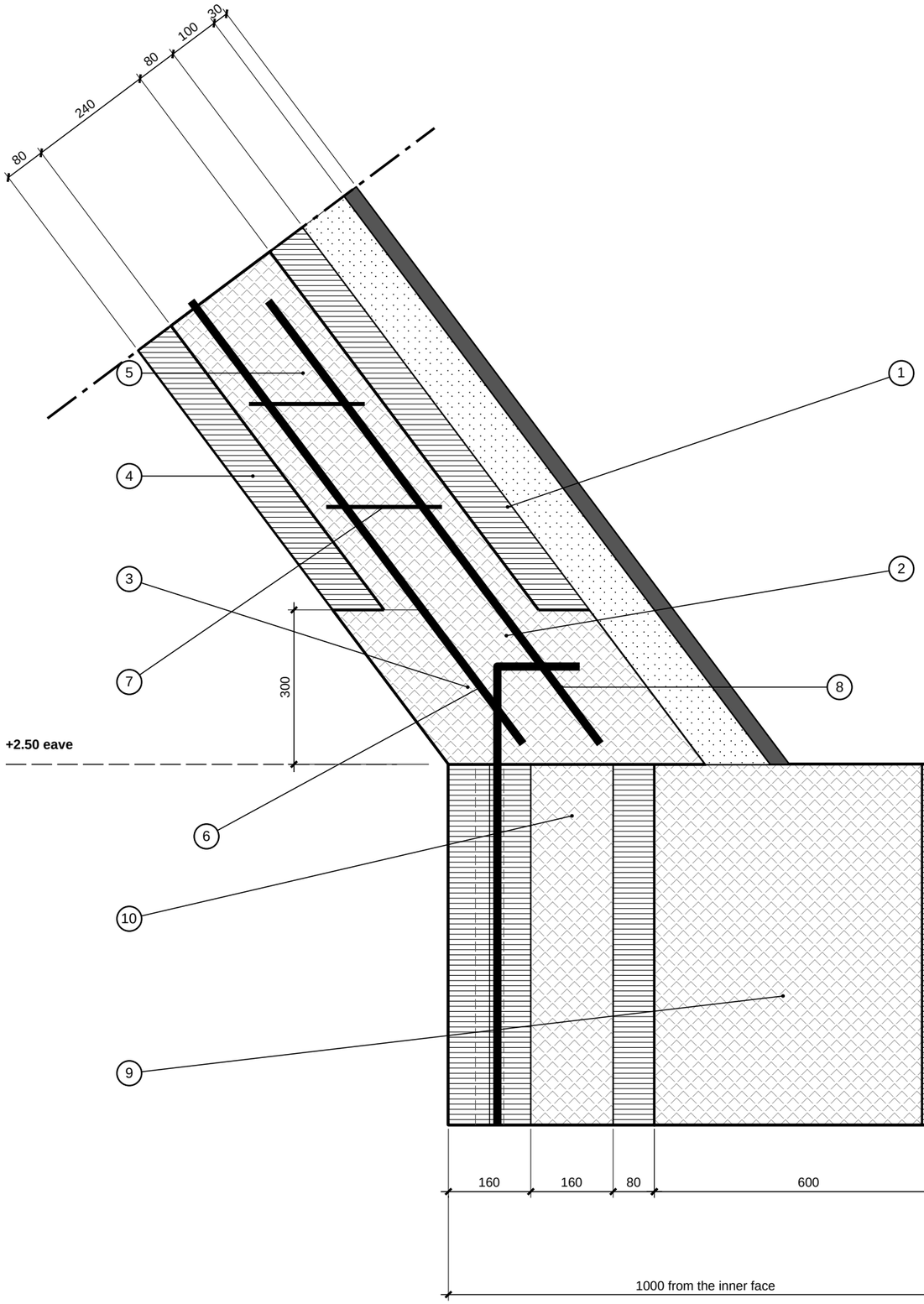
0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP

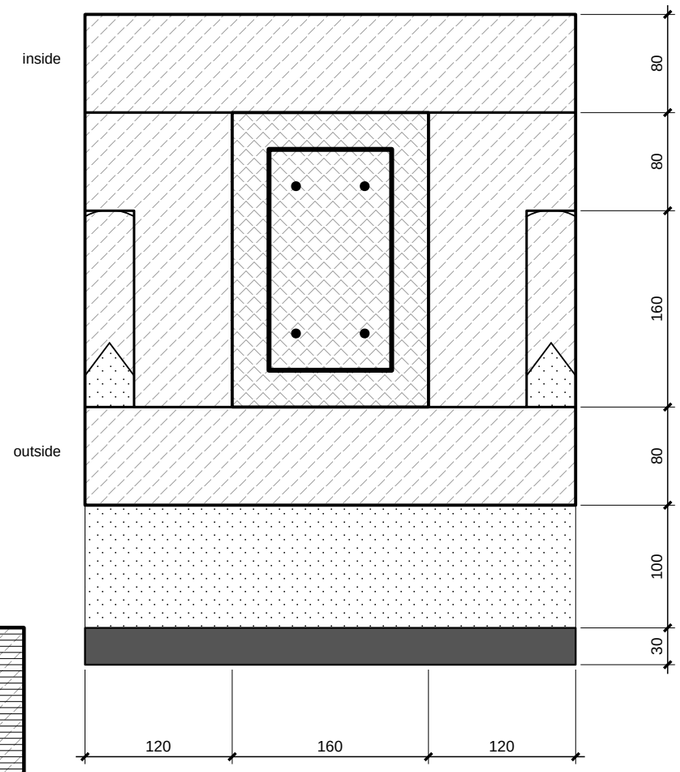
Scale 1:50 at A3 · rib plan cut 1:10, rib section 1:20 · dimensions in mm

2026-10-14 · hand estimates, not engineered

09



Plan cut through a rib at one layer, 1:5



Detail D1: eave, roof to wall

1. Roof outer layer 80, printed with the rest of the piece; outside it the insulation 100 (dotted) and the outer layer 30 (dark).
2. Zig-zag lattice 80 in a 160 zone; it touches the outer layer and layer 2 every @ 600; the cells between are open air pockets for ventilation.
3. Layer 2, 80: printed bead; beside each rib it turns into the printed channel walls (see the plan cut).
4. Inner layer 1, 80, room side. Layers 1 and 2 together are the structural leaf: 160 at the base, 80 at the crown (layer 2 ends).
5. Rib (cast column): channel 160 wide x 240 deep through layer 2 and the 160 zone, filled with poured concrete; section cut along the rib. It rises out of the solid base in the same pour, without a joint.
6. Rib bars R, 4 Ø16 (2 in the cut), anchored in the solid base where they cross the hooks of the wall bars A (assumed; steel fibre in the concrete could replace them).
7. Stirrups Ø8 every 200 (assumed).
8. Solid base of the roof: the first 300 (vertical) above the eave is full cast concrete over the whole 400, no air zone, poured in one go with the rib above it; wall bars A hook into it. Replaces the ring beam.
9. Buttress 400 wide, 1000 from the inner face (600 beyond the wall): printed formwork (80 end cap shown) filled with cast concrete; top flush with the roof base at +2.50.
10. Wall 400 at the buttress: printed inner layers and outer layer, the 160 zone between them cast and one body with the buttress (no insulation here, the cut is through the buttress, not the curtain wall). Wall bars A Ø16 run from the footing up into the solid base and end in a 160 hook (assumed). The roof grid is A or B (see the roof-grid sheets); the detail is cut along a rib.

0 1 2 3 4 dm

Pyramid roof · 3D-printed concrete · GSAPP
Scale 1:10 at A3 · rib plan cut 1:5 · dimensions in mm
2026-10-14 · hand estimates, not engineered